Multiplication of integers

1 Are the following statements true or false?

a The product of a positve number with a negative number is negative.

b The product of an even number of negative numbers is negative.

2 State whether the following expressions will be positive or negative:

a $(-147) \times 183$

b $(-151) \times (-185)$

c $(-180) \times (-168)$

d $157 \times (-142)$

3 Consider the equation $6 \times 4 = 24$, and find the value of the following:

a $(-6) \times 4$

b $6 \times (-4)$

c $(-6) \times (-4)$

4 Consider the equation $19 \times 17 = 323$, and find the value of the following:

a $(-19) \times 17$

b $19 \times (-17)$

c $(-19) \times (-17)$

5 Consider the equation $37 \times 31 = 1147$, and find the value of the following:

a $(-37) \times 31$

b $37 \times (-31)$

c $(-37) \times (-31)$

6 Consider the equation $3 \times 2 \times 5 = 30$, and find the value of the following:

a $3 \times (-2) \times 5$

b $(-3) \times 2 \times (-5)$

c $(-3) \times (-2) \times 5$

d $(-3) \times (-2) \times (-5)$

7 Evaluate:

a -4^2

b $(-4)^2$

c $(-4)^3$

d $(-12)^2$

e $-(-12)^2$

f $-(-8)^3$

g $(-6)^3$

h $6 \times (-6)$

i $5 \times (-9)$

j $(-12) \times (-10)$

k $(-9) \times 1$

l $(-9) \times 0$

m $(-11) \times 4$

n $(-4) \times (-11)$

o $(-4) \times (-4)$

p $(-4) \times (-4) \times (-4)$

q $4 \times (-7) \times 5$

r $(-3) \times 4 \times (-3)$

© ClipSAT: By Mr. Mohamed Abdallah +977597688647

8 Solve the following equations:

a $(-11) \times r = -66$

b $n \times (-3) = -36$

c $7 \times q = -28$

d $t \times 7 = -42$

9 Complete the following equations:

a $\square \times (-5) = 55$

b $-6 \times \square = -66$

c $\square \times 8 = -56$

d $7 \times \square = -84$

10 Find the value of n in the following statements:

a A number n is multiplied with -3 to give 21.

b A number n is multiplied with -10 to give -40.

11 Explain why there is no solution to the equation $x^2 = -1$.

Division of integers

12 Is the quotient of a positive integer and a negative integer positive or negative?

13 State whether the following expressions will be positive or negative:

a $(-2432) \div 32$

b $(-855) \div (-15)$

c $1587 \div (-23)$

d $(-7614) \div (-81)$

14 Consider the equation $42 \div 7 = 6$, and find the value of the following:

a $(-42) \div 7$

b $42 \div (-7)$

c $(-42) \div (-7)$

15 Consider the equation $667 \div 29 = 23$, and find the value of the following:

a $(-667) \div 29$

b $-667 \div (-29)$

c $(-667) \div (-29)$

16 Consider the equation $924 \div 28 = 33$, and find the value of the following:

a $(-924) \div 28$

b $924 \div (-28)$

c $(-924) \div (-28)$

© ClipSAT: By Mr. Mohamed Abdallah +977597688647

17 Evaluate:

a $(-2) \div (-2)$ b $(-2) \div 1$ c $4 \div (-1)$ d $0 \div (-36)$

e $12 \div (-4)$ f $(-20) \div 4$ g $21 \div (-7)$ h $(-22) \div 11$

i $(-50) \div (-10)$ j $(-72) \div (-9)$ k $\dfrac{-5}{5}$ l $\dfrac{-11}{-11}$

m $\dfrac{10}{-1}$ n $\dfrac{42}{-7}$ o $\dfrac{-60}{10}$ p $\dfrac{-48}{-12}$

18 Solve the following equations:

a $(-108) \div n = -12$ b $p \div (-8) = 11$

c $(-48) \div t = 6$ d $k \div 8 = -6$

19 Complete the following equations:

a $\square \div (-6) = -9$ b $\square \div 5 = -7$ c $\square \div 5 = -11$ d $(-72) \div \square = 9$

20 Find the value of n in the following statements:

a A number n, when divided by 3, gives -9.

b A number n when divided by -5 gives -4.

21 Evaluate:

a $(-4) \times 3 \div (-2)$ b $5 \times (-3) \times (-7)$

c $10 \times 8 \div (-5)$ d $(-6) \times (-4) \times (-3)$

e $(-14) \times (-4) \div 7$ f $8 \times (-15) \div 4$

g $(18 \div (-3)) \div (-2)$ h $(-126) \div ((-6) \times (-3))$

© ClipSAT: By Mr. Mohamed Abdallah +977597688647

Applications

22 Which quadrant(s) could the point (x, y) lie in if:

a $xy > 0$

b $\frac{x^2}{y} < 0$

23 In golf your score for each hole is determined by comparing it to the par (or expected) number of shots for that hole. Taking one less shot than expected will give a score of -1, one more shot will give $+1$, and so on.

Ivan played 18 holes of golf, and the frequency of each score he obtained is shown in the adjacent table:

Score	Frequency
-1	5
Par	3
$+1$	4
$+2$	6

a Find Ivan's total score for his 6 scores of $+2$.

b Find Ivan's total score for his 5 scores of -1.

c Find Ivan's total score for the total 18 holes.

24 Victoria earns $\$1100$ a month, and deposits it into her bank account. She also withdraws $\$200$ a month to pay for rent.

a How much, in total, does Victoria deposit into her bank account in one full year?

b How much, in total, does Victoria withdraw from her bank account in one full year?

c Find the total change in her bank account over the year. An increase will be represented by a positive number and a decrease will be represented by a negative number.

© ClipSAT: By Mr. Mohamed Abdallah +977597688647